

# Erich Robbi

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## Education

### PhD, Computer Science

UNIVERSITY OF TRENTO

- Research focus: **Hybrid decision-making** systems for high-stakes surgical interventions

Trento, Italy

Nov. 2023 - Current

### M.Sc., Data Science

UNIVERSITY OF TRENTO, GRADE 110/110 CUM LAUDE

- Thesis: AI-enhanced decision support for abdominal aortic aneurysm surgery, under prof. Andrea Passerini & prof. Marco Roveri

Trento, Italy

Sep. 2021 - Sep. 2023

### B.Sc., Interfaces and Communication Technologies

UNIVERSITY OF TRENTO, GRADE 105/110

Trento, Italy

Sep. 2018 - Sep. 2021

## Experience

### Researcher - Horizon Europe Project TANGO

UNIVERSITY OF TRENTO

- Leading a case-study for the **Horizon Europe TANGO project**
- Gathering requirements and iterating on feedback from **clinical end-users** for **high-stakes surgical planning**
- Managed collaborative workflows across a multidisciplinary team

Trento, Italy

Nov. 2023 - Current

### Lecturer

UNIVERSITY OF TRENTO

- Lecturer for the interdisciplinary course **Scientific Programming in Python** and **Data Structures and Algorithm**
- Mentored students in writing **efficient code** and **software development best practices**

Trento, Italy

Aug. 2025 - Current

### Junior Researcher

BRUNO KESSLER FOUNDATION

- Built **diffusion models** for histopathology image synthesis; validated with **expert-in-the-loop evaluations**

Trento, Italy

Jan. 2023 - Dec. 2023

## Projects

### End-to-End Surgical Planning Pipeline

- Built a **full-stack surgical planning platform** processing large CTA medical datasets for clinicians
- Designed and validated a **deep encoder-decoder segmentation model** to segment critical vascular anatomical regions in complex CTA scans, achieving state-of-the-art performance

### Expert-in-the-Loop Histopathology Generation

- Implemented a **WSI preprocessing pipeline**; processed TBs for **tile-level anomaly detection** to improve dataset quality for generative pathology models
- Trained and evaluated **deep generative models** for **real vs. synthetic** and **tissue detection** tasks to assess the fidelity of generated histopathology data
- Designed a **blinded evaluation framework** for pathologists, generating structured questionnaires to benchmark **AI model performance against expert human assessment**

### Explainable AI for Personalized Recommendations

- Developed a **personalized recommendation module** for a food e-commerce platform, generating optimized product bundles from **user profiles**
- Implemented a **preference elicitation** algorithm using the Choquet integral to model **non-additive preferences and multi-criteria decision making**
- Integrated Explainable AI techniques to produce **human-interpretable explanations** alongside recommendations, improving transparency and user trust

## Activities

### Mentor

MEDITECH CHALLENGE 2026

- Mentored a team of Master's and PhD students in the development of an **AI system for post-EVAR CT scan analysis**, supporting the project from problem formulation to technical validation; team received an **Honorary Mention**
- Guided the design and implementation of machine learning pipelines for **automated anatomical landmark detection** and **stent position assessment**

Trento, Italy

Feb. 2026

### Reviewer

- Served as reviewer for the following top-tier conferences: **MICCAI**, **IEEE: WACV**

Contributor

OPEN SOURCE CONTRIBUTION

- Developed a customizable **tile extraction module** enabling histopathology tiles to be selected using **user-defined scoring metrics**
- Extended **HistoLab** to support more flexible **whole slide image tile sampling** for computational pathology pipelines
- Added **unit tests** and updated project **documentation**; contribution integrated through a reviewed **pull request**

Oral Presentation

SIPMEL NATIONAL CONGRESS

- Presented a real-world case study demonstrating the integration of an **AI-based prediction model** into laboratory medicine workflows to support **clinical decision-making**
- Showcased how **machine learning** techniques can enhance diagnostic precision, reduce turnaround times, and support **evidence-based clinical judgments**

Remote

2026

Riva del Garda, Italy

Oct. 2023

Selected Posters / Publications

Contrast Arterial Transit Time on Final DSA Predicts Persistent Type II Endoleak after EVAR: A Pilot Study

AMERICAN ASSOCIATION OF PHYSICISTS IN MEDICINE - 2026

Vancouver, Canada

July - 2026

AI-Based Automated Extraction of Manufacturer-Specific Stent-Graft Codes for EVAR Planning

AMERICAN ASSOCIATION OF PHYSICISTS IN MEDICINE - 2026

Vancouver, Canada

July - 2026

Trust the Prior (or Not): Uncertainty-Aware Abdominal Aortic Aneurysm Segmentation

UNDER REVIEW

2026

Automated C-arm Angulations for Endovascular Aneurysm Repair

39TH EUROPEAN SOCIETY FOR VASCULAR SURGERY ANNUAL MEETING

Istanbul, Türkiye

Sep. 2025

The Role of AI-Based Analysis in Segmenting Sealing Zones and Tissue Characterization

AMERICAN ASSOCIATION OF PHYSICISTS IN MEDICINE - 2025

Washington D.C., USA

Jul. 2025

Automatic CTA analysis for blood vessels and aneurysm features extraction in EVAR planning

SCIENTIFIC REPORTS

2025

Validation of an Artificial Intelligence-Based Software for Automatic Segmentation and Endovascular Aneurysm Repair Planning in Patients with Abdominal Aortic Aneurysms

VEITHSYMPOSIUM - 2024

New York City, USA

Nov. 2024

AI-based Solutions for Better Endovascular Repair Strategies

HYBRID HUMAN-MACHINE INTERACTION IN SURGERY WORKSHOP AT HAMLYN SYMPOSIUM - 2024

London, United Kingdom

Jun. 2024

Generating and evaluating synthetic data in digital pathology through diffusion models

SCIENTIFIC REPORTS

2024

Personalized bundle recommendation using preference elicitation and the Choquet integral

FRONTIERS IN ARTIFICIAL INTELLIGENCE

2024

Skills

|                          |                                                                                       |
|--------------------------|---------------------------------------------------------------------------------------|
| ML / Research            | Machine Learning, Statistics, Quantitative Research, Medical Imaging, Computer Vision |
| Frameworks and Libraries | Pytorch, JAX, Pandas, NumPy, FastAPI, MONAI                                           |
| Programming              | Python (mainly), Javascript, Go, SQL, R, C#                                           |
| Tools                    | Linux, Docker, Git, VS Code, Slack, $\LaTeX$                                          |
| Languages                | Italian (native), English, Spanish, German                                            |